

9 THE WKB APPROXIMATION

The **WKB** (Wentzel, Kramers, Brillouin)¹ method is a technique for obtaining approximate solutions to the time-independent Schrödinger equation in one dimension (the same basic idea can be applied to many other differential equations, and to the radial part of the Schrödinger equation in three dimensions). It is particularly useful in calculating bound state energies and tunneling rates through potential barriers.

The essential idea is as follows: Imagine a particle of energy E moving through a region where the potential $V(x)$ is *constant*. If $E > V$, the wave function is of the form

$$\psi(x) = Ae^{\pm ikx}, \quad \text{with} \quad k \equiv \sqrt{2m(E - V)}/\hbar.$$

The plus sign indicates that the particle is traveling to the right, and the minus sign means it is going to the left (the general solution, of course, is a linear combination of the two). The wave function is oscillatory, with fixed wavelength ($\lambda = 2\pi/k$) and unchanging amplitude (A). Now suppose that $V(x)$ is *not* constant, but varies rather slowly in comparison to λ , so that over a region containing many full wavelengths the potential is *essentially* constant. Then it is reasonable to suppose that ψ remains *practically* sinusoidal, except that the wavelength and the amplitude change slowly with x . This is the inspiration behind the WKB approximation. In effect, it identifies two different levels of x -dependence: rapid oscillations, *modulated* by gradual variation in amplitude and wavelength.

By the same token, if $E < V$ (and V is constant), then ψ is exponential:

$$\psi(x) = Ae^{\pm \kappa x}, \quad \text{with} \quad \kappa \equiv \sqrt{2m(V - E)}/\hbar.$$

And if $V(x)$ is *not* constant, but varies slowly in comparison with $1/\kappa$, the solution remains *practically* exponential, except that A and κ are now slowly-varying functions of x .

Now, there is one place where this whole program is bound to fail, and that is in the immediate vicinity of a classical **turning point**, where $E \approx V$. For here λ (or $1/\kappa$) goes to infinity, and $V(x)$ can hardly be said to vary “slowly” in comparison. As we shall see, a proper handling of the turning points is the most difficult aspect of the WKB approximation, though the final results are simple to state and easy to implement.

9.1 THE “CLASSICAL” REGION

The Schrödinger equation,

$$-\frac{\hbar^2}{2m} \frac{d^2\psi}{dx^2} + V(x)\psi = E\psi,$$

¹ In Holland it's KWB, in France it's BWK, and in England it's JWKB (for Jeffreys).

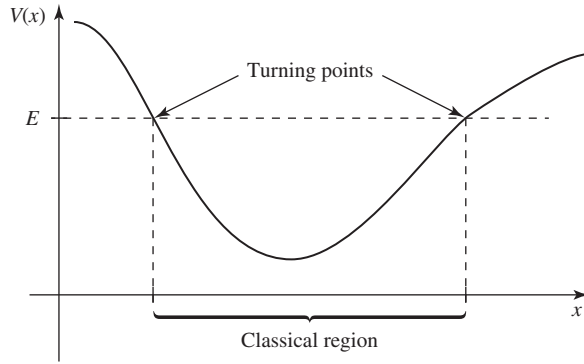


Figure 9.1: Classically, the particle is confined to the region where $E \geq V(x)$.

can be rewritten in the following way:

$$\frac{d^2\psi}{dx^2} = -\frac{p^2}{\hbar^2}\psi, \quad (9.1)$$

where

$$p(x) \equiv \sqrt{2m[E - V(x)]} \quad (9.2)$$

is the classical formula for the (magnitude of the) momentum of a particle with total energy E and potential energy $V(x)$. For the moment, I'll assume that $E > V(x)$, so that $p(x)$ is *real*; we call this the “classical” region, for obvious reasons—classically the particle is *confined* to this range of x (see Figure 9.1). In general, ψ is some complex function; we can express it in terms of its *amplitude*, $A(x)$, and its *phase*, $\phi(x)$ —both of which are *real*:

$$\psi(x) = A(x)e^{i\phi(x)}. \quad (9.3)$$

Using a prime to denote the derivative with respect to x ,

$$\frac{d\psi}{dx} = (A' + iA\phi')e^{i\phi},$$

and

$$\frac{d^2\psi}{dx^2} = [A'' + 2iA'\phi' + iA\phi'' - A(\phi')^2]e^{i\phi}. \quad (9.4)$$

Putting this into Equation 9.1:

$$A'' + 2iA'\phi' + iA\phi'' - A(\phi')^2 = -\frac{p^2}{\hbar^2}A. \quad (9.5)$$

This is equivalent to two *real* equations, one for the real part and one for the imaginary part:

$$A'' - A(\phi')^2 = -\frac{p^2}{\hbar^2}A, \quad \text{or} \quad A'' = A\left[(\phi')^2 - \frac{p^2}{\hbar^2}\right], \quad (9.6)$$

and

$$2A'\phi' + A\phi'' = 0, \quad \text{or} \quad (A^2\phi')' = 0. \quad (9.7)$$

Equations 9.6 and 9.7 are entirely equivalent to the original Schrödinger equation. The second one is easily solved:

$$A^2 \phi' = C^2, \quad \text{or} \quad A = \frac{C}{\sqrt{|\phi'|}}, \quad (9.8)$$

where C is a (real) constant. The first one (Equation 9.6) cannot be solved in general—so here comes the approximation: *We assume that the amplitude A varies slowly*, so the A'' term is negligible. (More precisely, we assume that A''/A is much less than both $(\phi')^2$ and $p^2/\hbar^2$.) In that case we can drop the left side of Equation 9.6, and we are left with

$$(\phi')^2 = \frac{p^2}{\hbar^2}, \quad \text{or} \quad \frac{d\phi}{dx} = \pm \frac{p}{\hbar},$$

and therefore

$$\phi(x) = \pm \frac{1}{\hbar} \int p(x) dx. \quad (9.9)$$

(I'll write this as an *indefinite* integral, for now—any constant of integration can be absorbed into C , which thereby becomes complex. I'll also absorb a factor of $\sqrt{\hbar}$.) Then

$$\psi(x) \approx \frac{C}{\sqrt{p(x)}} e^{\pm \frac{i}{\hbar} \int p(x) dx}. \quad (9.10)$$

Notice that

$$|\psi(x)|^2 \approx \frac{|C|^2}{p(x)}, \quad (9.11)$$

which says that the probability of finding the particle at point x is inversely proportional to its (classical) momentum (and hence its velocity) at that point. This is exactly what you would expect—the particle doesn't spend long in the places where it is moving rapidly, so the probability of getting caught there is small. In fact, the WKB approximation is sometimes *derived* by starting with this “semi-classical” observation, instead of by dropping the A'' term in the differential equation. The latter approach is cleaner mathematically, but the former offers a more illuminating physical rationale. The general (approximate) solution, of course, will be a linear combination the two solutions in Equation 9.10, one with each sign.

Example 9.1

Potential well with two vertical walls. Suppose we have an infinite square well with a bumpy bottom (Figure 9.2):

$$V(x) = \begin{cases} \text{some specified function,} & (0 < x < a), \\ \infty, & (\text{otherwise}). \end{cases} \quad (9.12)$$

Inside the well (assuming $E > V(x)$ throughout) we have

$$\psi(x) \approx \frac{1}{\sqrt{p(x)}} [C_+ e^{i\phi(x)} + C_- e^{-i\phi(x)}], \quad (9.13)$$

or, more conveniently,

$$\psi(x) \approx \frac{1}{\sqrt{p(x)}} [C_1 \sin \phi(x) + C_2 \cos \phi(x)], \quad (9.14)$$

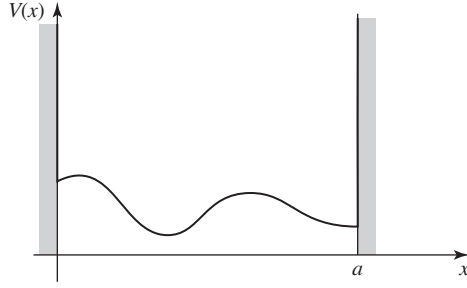


Figure 9.2: Infinite square well with a bumpy bottom.

where (exploiting the freedom noted earlier to impose a convenient lower limit on the integral)²

$$\phi(x) = \frac{1}{\hbar} \int_0^x p(x') dx'. \quad (9.15)$$

Now, $\psi(x)$ must go to zero at $x = 0$, and therefore (since $\phi(0) = 0$) $C_2 = 0$. Also, $\psi(x)$ goes to zero at $x = a$, so

$$\phi(a) = n\pi \quad (n = 1, 2, 3, \dots). \quad (9.16)$$

Conclusion:

$$\boxed{\int_0^a p(x) dx = n\pi\hbar.} \quad (9.17)$$

This quantization condition determines the (approximate) allowed energies.

For instance, if the well has a *flat* bottom ($V(x) = 0$), then $p(x) = \sqrt{2mE}$ (a constant), and Equation 9.17 says $pa = n\pi\hbar$, or

$$E_n = \frac{n^2 \pi^2 \hbar^2}{2ma^2},$$

which is the old formula for the energy levels of the infinite square well (Equation 2.30). In this case the WKB approximation yields the *exact* answer (the amplitude of the true wave function is *constant*, so dropping A'' cost us nothing).

* **Problem 9.1** Use the WKB approximation to find the allowed energies (E_n) of an infinite square well with a “shelf,” of height V_0 , extending half-way across (Figure 7.3):

$$V(x) = \begin{cases} V_0, & (0 < x < a/2), \\ 0, & (a/2 < x < a), \\ \infty, & (\text{otherwise}). \end{cases}$$

² We might as well take the positive sign, since both are covered by Equation 9.13.

Express your answer in terms of V_0 and $E_n^0 \equiv (n\pi\hbar)^2/2ma^2$ (the n th allowed energy for the infinite square well with *no* shelf). Assume that $E_1^0 > V_0$, but do *not* assume that $E_n \gg V_0$. Compare your result with what we got in Section 7.1.2, using first-order perturbation theory. Note that they are in agreement if either V_0 is very small (the perturbation theory regime) or n is very large (the WKB—semi-classical—regime).

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Problem 9.2 An alternative derivation of the WKB formula (Equation 9.10) is based on an expansion in powers of $\hbar$. Motivated by the free-particle wave function, $\psi = A \exp(\pm ipx/\hbar)$, we write

$$\psi(x) = e^{if(x)/\hbar},$$

where $f(x)$ is some complex function. (Note that there is no loss of generality here—any nonzero function can be written in this way.)

(a) Put this into Schrödinger's equation (in the form of Equation 9.1), and show that

$$i\hbar f'' - (f')^2 + p^2 = 0.$$

(b) Write $f(x)$ as a power series in $\hbar$:

$$f(x) = f_0(x) + \hbar f_1(x) + \hbar^2 f_2(x) + \cdots,$$

and, collecting like powers of $\hbar$, show that

$$(f_0')^2 = p^2, \quad if_0'' = 2f_0'f_1', \quad if_1'' = 2f_0'f_2' + (f_1')^2, \quad \text{etc.}$$

(c) Solve for $f_0(x)$ and $f_1(x)$, and show that—to first order in $\hbar$ —you recover Equation 9.10.

Note: The logarithm of a negative number is defined by $\ln(-z) = \ln(z) + in\pi$, where n is an odd integer. If this formula is new to you, try exponentiating both sides, and you'll see where it comes from.

9.2 TUNNELING

So far, I have assumed that $E > V$, so $p(x)$ is real. But we can easily write down the corresponding result in the *non*-classical region ($E < V$)—it's the same as before (Equation 9.10), only now $p(x)$ is *imaginary*:³

$$\psi(x) \approx \frac{C}{\sqrt{|p(x)|}} e^{\pm \frac{1}{\hbar} \int |p(x)| dx}. \quad (9.18)$$

Consider, for example, the problem of scattering from a rectangular barrier with a bumpy top (Figure 9.3). To the left of the barrier ($x < 0$),

$$\psi(x) = Ae^{ikx} + Be^{-ikx}, \quad (9.19)$$

³ In this case the wave function is *real*, and the analogs to Equations 9.6 and 9.7 do not follow *necessarily* from Equation 9.5, although they are still *sufficient*. If this bothers you, study the alternative derivation in Problem 9.2.

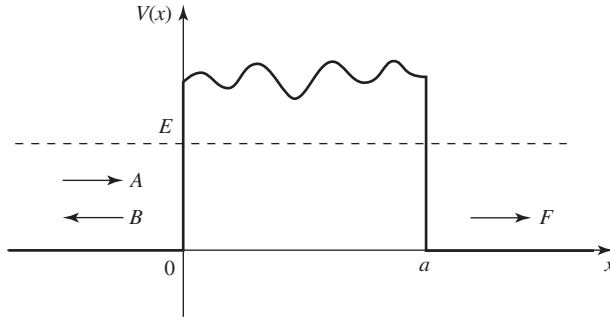


Figure 9.3: Scattering from a rectangular barrier with a bumpy top.

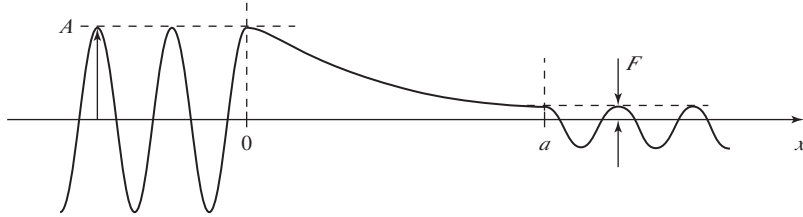


Figure 9.4: Qualitative structure of the wave function, for scattering from a high, broad barrier.

where A is the incident amplitude, B is the reflected amplitude, and $k \equiv \sqrt{2mE}/\hbar$ (see Section 2.5). To the right of the barrier ($x > a$),

$$\psi(x) = F e^{ikx}, \quad (9.20)$$

where F is the transmitted amplitude. The transmission probability is

$$T = \frac{|F|^2}{|A|^2}. \quad (9.21)$$

In the tunneling region ($0 \leq x \leq a$), the WKB approximation gives

$$\psi(x) \approx \frac{C}{\sqrt{|p(x)|}} e^{\frac{1}{\hbar} \int_0^x |p(x')| dx'} + \frac{D}{\sqrt{|p(x)|}} e^{-\frac{1}{\hbar} \int_0^x |p(x')| dx'}. \quad (9.22)$$

If the barrier is very high and/or very wide (which is to say, if the probability of tunneling is small), then the coefficient of the exponentially *increasing* term (C) must be small (in fact, it would be *zero* if the barrier were *infinitely* broad), and the wave function looks something like⁴ Figure 9.4. The relative amplitudes of the incident and transmitted waves are determined essentially by the total decrease of the exponential over the nonclassical region:

$$\frac{|F|}{|A|} \sim e^{-\frac{1}{\hbar} \int_0^a |p(x')| dx'},$$

so

$$T \approx e^{-2\gamma}, \quad \text{with} \quad \gamma \equiv \frac{1}{\hbar} \int_0^a |p(x)| dx. \quad (9.23)$$

⁴ This heuristic argument can be made more rigorous—see Problem 9.11.

Example 9.2

Gamow's theory of alpha decay.⁵ In 1928, George Gamow (and, independently, Condon and Gurney) used Equation 9.23 to provide the first successful explanation of alpha decay (the spontaneous emission of an alpha particle—two protons and two neutrons—by certain radioactive nuclei).⁶ Since the alpha particle carries a positive charge ($2e$), it will be electrically repelled by the leftover nucleus (charge Ze), as soon as it gets far enough away to escape the nuclear binding force. But first it has to negotiate a potential barrier that was already known (in the case of uranium) to be more than twice the energy of the emitted alpha particle. Gamow approximated the potential energy by a finite square well (representing the attractive nuclear force), extending out to r_1 (the radius of the nucleus), joined to a repulsive Coulombic tail (Figure 9.5), and identified the escape mechanism as quantum tunneling (this was, by the way, the first time that quantum mechanics had been applied to nuclear physics).

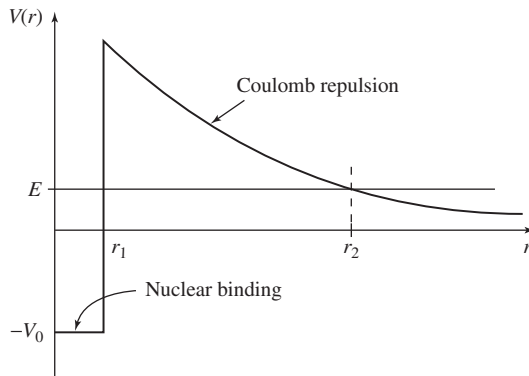


Figure 9.5: Gamow's model for the potential energy of an alpha particle in a radioactive nucleus.

If E is the energy of the emitted alpha particle, the outer turning point (r_2) is determined by

$$\frac{1}{4\pi\epsilon_0} \frac{2Ze^2}{r_2} = E. \quad (9.24)$$

The exponent γ (Equation 9.23) is evidently⁷

$$\gamma = \frac{1}{\hbar} \int_{r_1}^{r_2} \sqrt{2m \left(\frac{1}{4\pi\epsilon_0} \frac{2Ze^2}{r} - E \right)} dr = \frac{\sqrt{2mE}}{\hbar} \int_{r_1}^{r_2} \sqrt{\frac{r_2}{r} - 1} dr.$$

The integral can be done by substitution (let $r \equiv r_2 \sin^2 u$), and the result is

$$\gamma = \frac{\sqrt{2mE}}{\hbar} \left[r_2 \left(\frac{\pi}{2} - \sin^{-1} \sqrt{\frac{r_1}{r_2}} \right) - \sqrt{r_1(r_2 - r_1)} \right]. \quad (9.25)$$

⁵ For a more complete discussion, and alternative formulations, see B. R. Holstein, *Am. J. Phys.* **64**, 1061 (1996).

⁶ For an interesting brief history see E. Merzbacher, "The Early History of Quantum Tunneling," *Physics Today*, August 2002, p. 44.

⁷ In this case the potential does not drop to zero on the left side of the barrier (moreover, this is really a three-dimensional problem), but the essential idea, contained in Equation 9.23, is all we really need.

Typically, $r_1 \ll r_2$, and we can simplify this result using the small angle approximation ($\sin \epsilon \approx \epsilon$):

$$\gamma \cong \frac{\sqrt{2mE}}{\hbar} \left[\frac{\pi}{2} r_2 - 2\sqrt{r_1 r_2} \right] = K_1 \frac{Z}{\sqrt{E}} - K_2 \sqrt{Z r_1}, \quad (9.26)$$

where

$$K_1 \equiv \left(\frac{e^2}{4\pi\epsilon_0} \right) \frac{\pi\sqrt{2m}}{\hbar} = 1.980 \text{ MeV}^{1/2}, \quad (9.27)$$

and

$$K_2 \equiv \left(\frac{e^2}{4\pi\epsilon_0} \right)^{1/2} \frac{4\sqrt{m}}{\hbar} = 1.485 \text{ fm}^{-1/2}. \quad (9.28)$$

(One fermi (fm) is 10^{-15} m, which is about the size of a typical nucleus.)

If we imagine the alpha particle rattling around inside the nucleus, with an average velocity v , the time between “collisions” with the “wall” is about $2r_1/v$, and hence the frequency of collisions is $v/2r_1$. The probability of escape at each collision is $e^{-2\gamma}$, so the probability of emission, per unit time, is $(v/2r_1) e^{-2\gamma}$, and hence the **lifetime** of the parent nucleus is about

$$\tau = \frac{2r_1}{v} e^{2\gamma}. \quad (9.29)$$

Unfortunately, we don't know v —but it hardly matters, for the exponential factor varies over a *fantastic* range (twenty-five orders of magnitude), as we go from one radioactive nucleus to another; relative to this the variation in v is pretty insignificant. In particular, if you plot the *logarithm* of the experimentally measured lifetime against $1/\sqrt{E}$, the result is a beautiful straight line (Figure 9.6),⁸ just as you would expect from Equations 9.26 and 9.29.

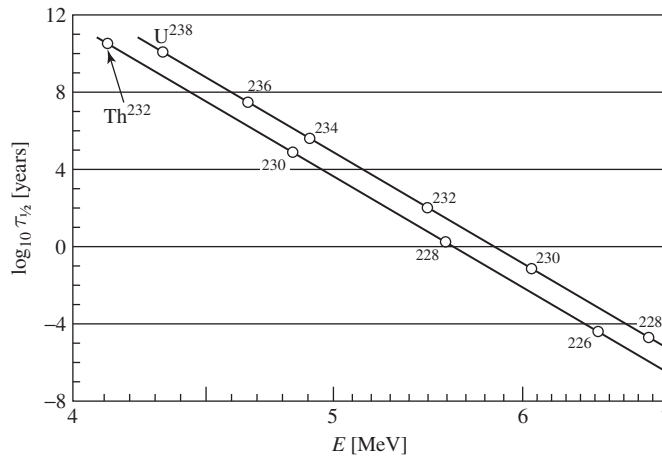


Figure 9.6: Graph of the logarithm of the half-life ($\tau_{1/2} = \tau \ln 2$) versus $1/\sqrt{E}$ (where E is the energy of the emitted alpha particle), for isotopes of uranium and thorium.

⁸ This figure is reprinted by permission from David Park, *Introduction to the Quantum Theory*, 3rd edn, Dover Publications, New York (2005); it was adapted from I. Perlman and J. O. Rasmussen, “Alpha Radioactivity,” *Encyclopedia of Physics*, Vol. 42, Springer (1957).

- * **Problem 9.3** Use Equation 9.23 to calculate the approximate transmission probability for a particle of energy E that encounters a finite square barrier of height $V_0 > E$ and width $2a$. Compare your answer with the exact result (Problem 2.33), to which it should reduce in the WKB regime $T \ll 1$.

- ** **Problem 9.4** Calculate the lifetimes of U^{238} and Po^{212} , using Equations 9.26 and 9.29. *Hint:* The density of nuclear matter is relatively constant (i.e. the same for all nuclei), so $(r_1)^3$ is proportional to A (the number of neutrons plus protons). Empirically,

$$r_1 \cong (1.07 \text{ fm}) A^{1/3}. \quad (9.30)$$

The energy of the emitted alpha particle can be deduced by using Einstein's formula ($E = mc^2$):

$$E = m_p c^2 - m_d c^2 - m_\alpha c^2, \quad (9.31)$$

where m_p is the mass of the parent nucleus, m_d is the mass of the daughter nucleus, and m_α is the mass of the alpha particle (which is to say, the He^4 nucleus). To figure out what the daughter nucleus is, note that the alpha particle carries off two protons and two neutrons, so Z decreases by 2 and A by 4. Look up the relevant nuclear masses. To estimate v , use $E = (1/2) m_\alpha v^2$; this ignores the (negative) potential energy inside the nucleus, and surely *underestimates* v , but it's about the best we can do at this stage. Incidentally, the experimental lifetimes are 6×10^9 yrs and $0.5 \mu\text{s}$, respectively.

Problem 9.5 Zener Tunneling. In a semiconductor, an electric field (if it's large enough) can produce transitions between energy bands—a phenomenon known as Zener tunneling. A uniform electric field $\mathbf{E} = -E_0 \hat{i}$, for which

$$H' = -e E_0 x,$$

makes the energy bands position dependent, as shown in Figure 9.7. It is then possible for an electron to tunnel from the valence (lower) band to the conduction (upper) band; this phenomenon is the basis for the **Zener diode**. Treating the gap as a potential barrier through which the electron may tunnel, find the tunneling probability in terms of E_g and E_0 (as well as m , $\hbar$, e).

9.3 THE CONNECTION FORMULAS

In the discussion so far I have assumed that the “walls” of the potential well (or the barrier) are *vertical*, so that the “exterior” solution is simple, and the boundary conditions trivial. As it turns out, our main results (Equations 9.17 and 9.23) are reasonably accurate even when the edges are not so abrupt (indeed, in Gamow's theory they were applied to just such a case). Nevertheless, it is of some interest to study more closely what happens to the wave function at a turning point ($E = V$), where the “classical” region joins the “nonclassical” region, and

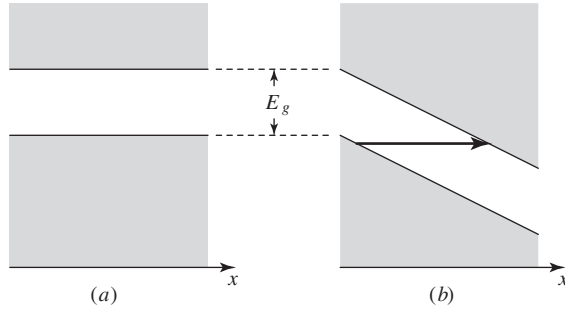


Figure 9.7: (a) The energy bands in the absence of an electric field. (b) In the presence of an electric field an electron can tunnel between the energy bands.

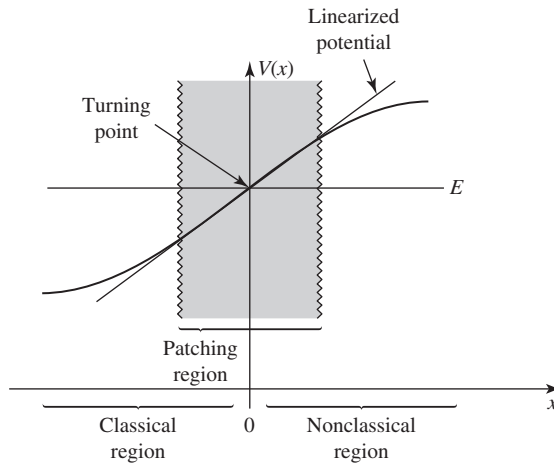


Figure 9.8: Enlarged view of the right-hand turning point.

the WKB approximation itself breaks down. In this section I'll treat the bound state problem (Figure 9.1); you get to do the scattering problem for yourself (Problem 9.11).⁹

For simplicity, let's shift the axes over so that the right hand turning point occurs at $x = 0$ (Figure 9.8). In the WKB approximation, we have

$$\psi(x) \approx \begin{cases} \frac{1}{\sqrt{p(x)}} \left[B e^{\frac{i}{\hbar} \int_x^0 p(x') dx'} + C e^{-\frac{i}{\hbar} \int_x^0 p(x') dx'} \right], & x < 0, \\ \frac{1}{\sqrt{|p(x)|}} D e^{-\frac{1}{\hbar} \int_0^x |p(x')| dx'}, & x > 0. \end{cases} \quad (9.32)$$

(Assuming $V(x)$ remains greater than E for all $x > 0$, we can exclude the positive exponent in this region, because it blows up as $x \rightarrow \infty$.) Our task is to join the two solutions at the boundary. But there is a serious difficulty here: In the WKB approximation, ψ goes to *infinity* at the turning point (where $p(x) \rightarrow 0$). The *true* wave function, of course, has no such wild behavior—as anticipated, the WKB method simply fails in the vicinity of a turning point. And yet, it is precisely the boundary conditions at the turning points that determine the

⁹ Warning: The following argument is quite technical, and you may wish to skip it on a first reading.

allowed energies. What we need to do, then, is *splice* the two WKB solutions together, using a “patching” wave function that straddles the turning point.

Since we only need the patching wave function (ψ_p) in the neighborhood of the origin, we’ll *approximate the potential by a straight line*:

$$V(x) \approx E + V'(0)x, \quad (9.33)$$

and solve the Schrödinger equation for this linearized V :

$$-\frac{\hbar^2}{2m} \frac{d^2\psi_p}{dx^2} + (E + V'(0)x)\psi_p = E\psi_p,$$

or

$$\frac{d^2\psi_p}{dx^2} = \alpha^3 x \psi_p, \quad (9.34)$$

where

$$\alpha \equiv \left[\frac{2m}{\hbar^2} V'(0) \right]^{1/3}. \quad (9.35)$$

The α s can be absorbed into the independent variable by defining

$$z \equiv \alpha x, \quad (9.36)$$

so that

$$\frac{d^2\psi_p}{dz^2} = z\psi_p. \quad (9.37)$$

This is **Airy’s equation**, and the solutions are called **Airy functions**.¹⁰ Since the Airy equation is a second-order differential equation, there are two linearly independent Airy functions, $\text{Ai}(z)$ and $\text{Bi}(z)$. They are related to Bessel functions of order $1/3$; some of their properties are listed in Table 9.1 and they are plotted in Figure 9.9. Evidently the patching wave function is a linear combination of $\text{Ai}(z)$ and $\text{Bi}(z)$:

$$\psi_p(x) = a\text{Ai}(\alpha x) + b\text{Bi}(\alpha x), \quad (9.38)$$

for appropriate constants a and b .

Now, ψ_p is the (approximate) wave function in the neighborhood of the origin; our job is to match it to the WKB solutions in the overlap regions on either side (see Figure 9.10). These overlap zones are close enough to the turning point that the linearized potential is reasonably accurate (so that ψ_p is a good approximation to the true wave function), and yet far enough away from the turning point that the WKB approximation is reliable.¹¹ In the overlap regions Equation 9.33 holds, and therefore (in the notation of Equation 9.35)

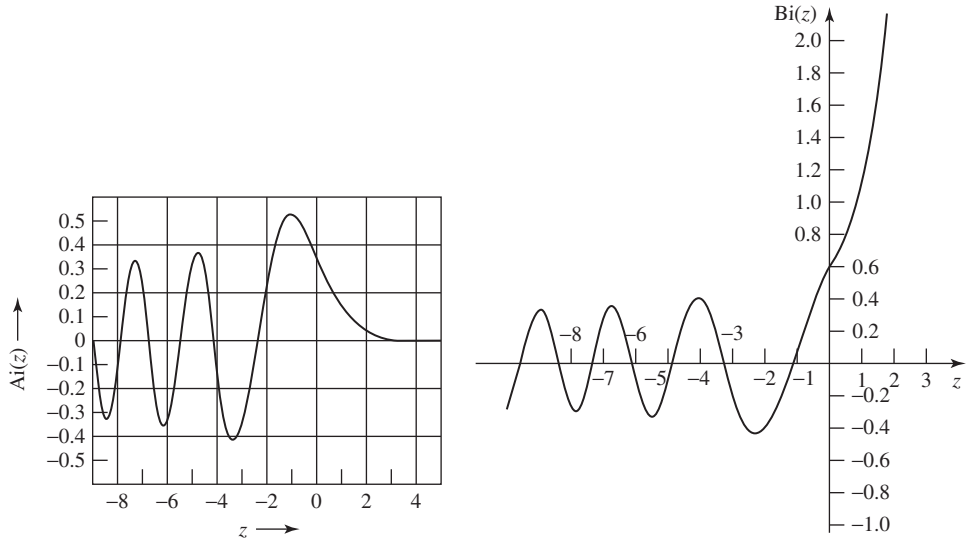
$$p(x) \approx \sqrt{2m(E - E - V'(0)x)} = \hbar\alpha^{3/2}\sqrt{-x}. \quad (9.39)$$

¹⁰ Classically, a linear potential means a constant force, and hence a constant acceleration—the simplest nontrivial motion possible, and the *starting* point for elementary mechanics. It is ironic that the same potential in *quantum* mechanics yields stationary states that are unfamiliar transcendental functions, and plays only a peripheral role in the theory. Still, wave *packets* can be reasonably simple—see Problem 2.51 and especially footnote 61, page 81.

¹¹ This is a delicate double constraint, and it is possible to concoct potentials so pathological that no such overlap region exists. However, in practical applications this seldom occurs. See Problem 9.9.

Table 9.1: *Some properties of the Airy functions.*

Differential Equation:	$\frac{d^2y}{dz^2} = zy.$
Solutions:	Linear combinations of Airy functions, $\text{Ai}(z)$ and $\text{Bi}(z)$.
Integral Representation:	$\text{Ai}(z) = \frac{1}{\pi} \int_0^\infty \cos\left(\frac{s^3}{3} + sz\right) ds,$ $\text{Bi}(z) = \frac{1}{\pi} \int_0^\infty \left[e^{-\frac{s^3}{3} + sz} + \sin\left(\frac{s^3}{3} + sz\right) \right] ds.$
Asymptotic Forms:	$\left. \begin{aligned} \text{Ai}(z) &\sim \frac{1}{2\sqrt{\pi}z^{1/4}} e^{-\frac{2}{3}z^{3/2}} \\ \text{Bi}(z) &\sim \frac{1}{\sqrt{\pi}z^{1/4}} e^{\frac{2}{3}z^{3/2}} \end{aligned} \right\} z \gg 0;$ $\left. \begin{aligned} \text{Ai}(z) &\sim \frac{1}{\sqrt{\pi}(-z)^{1/4}} \sin\left[\frac{2}{3}(-z)^{3/2} + \frac{\pi}{4}\right] \\ \text{Bi}(z) &\sim \frac{1}{\sqrt{\pi}(-z)^{1/4}} \cos\left[\frac{2}{3}(-z)^{3/2} + \frac{\pi}{4}\right] \end{aligned} \right\} z \ll 0.$

**Figure 9.9:** Graph of the Airy functions.

In particular, in overlap region 2,

$$\int_0^x |p(x')| dx' \approx \hbar \alpha^{3/2} \int_0^x \sqrt{x'} dx' = \frac{2}{3} \hbar (\alpha x)^{3/2},$$

and therefore the WKB wave function (Equation 9.32) can be written as

$$\psi(x) \approx \frac{D}{\sqrt{\hbar \alpha^{3/4} x^{1/4}}} e^{-\frac{2}{3}(\alpha x)^{3/2}}. \quad (9.40)$$

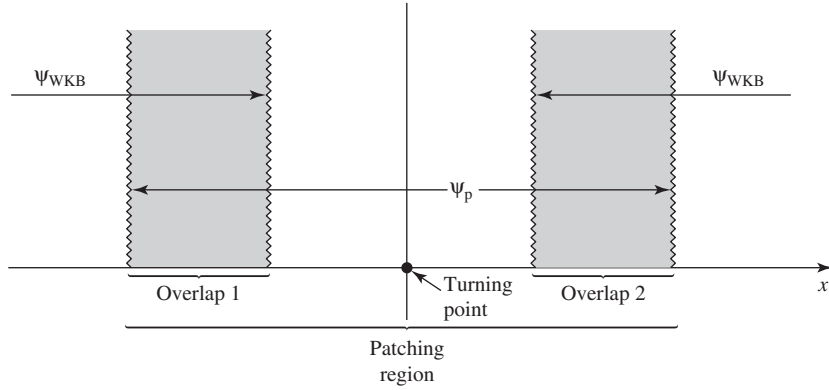


Figure 9.10: Patching region and the two overlap zones.

Meanwhile, using the large- z asymptotic forms¹² of the Airy functions (from Table 9.1), the patching wave function (Equation 9.38) in overlap region 2 becomes

$$\psi_p(x) \approx \frac{a}{2\sqrt{\pi}(\alpha x)^{1/4}} e^{-\frac{2}{3}(\alpha x)^{3/2}} + \frac{b}{\sqrt{\pi}(\alpha x)^{1/4}} e^{\frac{2}{3}(\alpha x)^{3/2}}. \quad (9.41)$$

Comparing the two solutions, we see that

$$a = \sqrt{\frac{4\pi}{\alpha\hbar}} D, \quad \text{and} \quad b = 0. \quad (9.42)$$

Now we go back and repeat the procedure for overlap region 1. Once again, $p(x)$ is given by Equation 9.39, but this time x is *negative*, so

$$\int_x^0 p(x') dx' \approx \frac{2}{3}\hbar(-\alpha x)^{3/2} \quad (9.43)$$

and the WKB wave function (Equation 9.32) is

$$\psi(x) \approx \frac{1}{\sqrt{\hbar\alpha^{3/4}(-x)^{1/4}}} \left[B e^{i\frac{2}{3}(-\alpha x)^{3/2}} + C e^{-i\frac{2}{3}(-\alpha x)^{3/2}} \right]. \quad (9.44)$$

Meanwhile, using the asymptotic form of the Airy function for large *negative* z (Table 9.1), the patching function (Equation 9.38, with $b = 0$) reads

$$\begin{aligned} \psi_p(x) &\approx \frac{a}{\sqrt{\pi}(-\alpha x)^{1/4}} \sin \left[\frac{2}{3}(-\alpha x)^{3/2} + \frac{\pi}{4} \right] \\ &= \frac{a}{\sqrt{\pi}(-\alpha x)^{1/4}} \frac{1}{2i} \left[e^{i\pi/4} e^{i\frac{2}{3}(-\alpha x)^{3/2}} - e^{-i\pi/4} e^{-i\frac{2}{3}(-\alpha x)^{3/2}} \right]. \end{aligned} \quad (9.45)$$

¹² At first glance it seems absurd to use a *large- z* approximation in this region, which after all is supposed to be reasonably close to the turning point at $z = 0$ (so that the linear approximation to the potential is valid). But notice that the argument here is αx , and if you study the matter carefully (see Problem 9.9) you will find that there *is* (typically) a region in which αx is large, but at the same time it is reasonable to approximate $V(x)$ by a straight line. Indeed, the asymptotic forms of the Airy functions *are* precisely the WKB solutions to Airy's equation, and since we are already using ψ_{WKB} in the overlap region (Figure 9.10) it is not really a *new* approximation to do the same for ψ_p .

Comparing the WKB and patching wave functions in overlap region 1, we find

$$\frac{a}{2i\sqrt{\pi}}e^{i\pi/4} = \frac{B}{\sqrt{\hbar\alpha}} \quad \text{and} \quad \frac{-a}{2i\sqrt{\pi}}e^{-i\pi/4} = \frac{C}{\sqrt{\hbar\alpha}},$$

or, putting in Equation 9.42 for a :

$$B = -ie^{i\pi/4}D, \quad \text{and} \quad C = ie^{-i\pi/4}D. \quad (9.46)$$

These are the so-called **connection formulas**, joining the WKB solutions at either side of the turning point. We're done with the patching wave function now—its only purpose was to bridge the gap. Expressing everything in terms of the one normalization constant D , and shifting the turning point back from the origin to an arbitrary point x_2 , the WKB wave function (Equation 9.32) becomes

$$\psi(x) \approx \begin{cases} \frac{2D}{\sqrt{p(x)}} \sin \left[\frac{1}{\hbar} \int_x^{x_2} p(x') dx' + \frac{\pi}{4} \right], & x < x_2; \\ \frac{D}{\sqrt{|p(x)|}} \exp \left[-\frac{1}{\hbar} \int_{x_2}^x |p(x')| dx' \right], & x > x_2. \end{cases} \quad (9.47)$$

Example 9.3

Potential well with one vertical wall. Imagine a potential well that has one vertical side (at $x = 0$) and one sloping side (Figure 9.11). In this case $\psi(0) = 0$, so Equation 9.47 says

$$\frac{1}{\hbar} \int_0^{x_2} p(x) dx + \frac{\pi}{4} = n\pi, \quad (n = 1, 2, 3, \dots),$$

or

$$\int_0^{x_2} p(x) dx = \left(n - \frac{1}{4} \right) \pi \hbar. \quad (9.48)$$

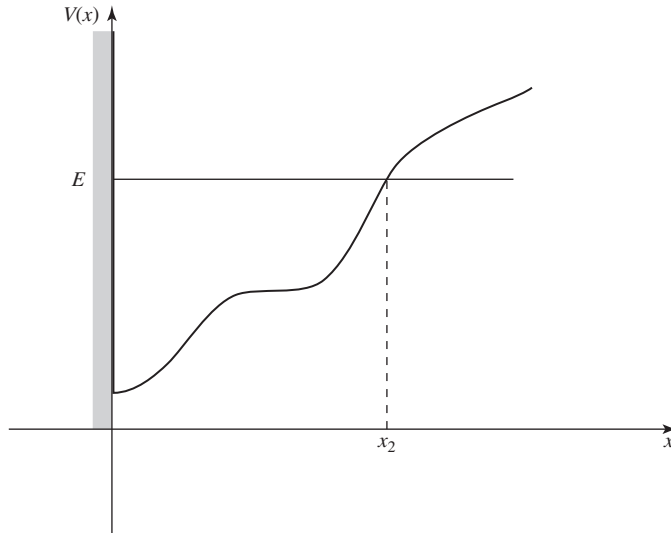


Figure 9.11: Potential well with one vertical wall.

For instance, consider the “half-harmonic oscillator”,

$$V(x) = \begin{cases} \frac{1}{2}m\omega^2 x^2, & (x > 0), \\ \infty, & (\text{otherwise}). \end{cases} \quad (9.49)$$

In this case

$$p(x) = \sqrt{2m[E - (1/2)m\omega^2 x^2]} = m\omega\sqrt{x_2^2 - x^2},$$

where

$$x_2 = \frac{1}{\omega}\sqrt{\frac{2E}{m}}$$

is the turning point. So

$$\int_0^{x_2} p(x) dx = m\omega \int_0^{x_2} \sqrt{x_2^2 - x^2} dx = \frac{\pi}{4}m\omega x_2^2 = \frac{\pi E}{2\omega},$$

and the quantization condition (Equation 9.48) yields

$$E_n = \left(2n - \frac{1}{2}\right) \hbar\omega = \left(\frac{3}{2}, \frac{7}{2}, \frac{11}{2}, \dots\right) \hbar\omega. \quad (9.50)$$

In this particular case the WKB approximation actually delivers the *exact* allowed energies (which are precisely the *odd* energies of the *full* harmonic oscillator—see Problem 2.41).

Example 9.4

Potential well with no vertical walls. Equation 9.47 connects the WKB wave functions at a turning point where the potential slopes *upward* (Figure 9.12(a)); the same reasoning, applied to a *downward*-sloping turning point (Figure 9.12(b)), yields (Problem 9.10)

$$\psi(x) \approx \begin{cases} \frac{D'}{\sqrt{|p(x)|}} \exp\left[-\frac{1}{\hbar} \int_x^{x_1} |p(x')| dx'\right], & x < x_1; \\ \frac{2D'}{\sqrt{p(x)}} \sin\left[\frac{1}{\hbar} \int_{x_1}^x p(x') dx' + \frac{\pi}{4}\right], & x > x_1. \end{cases} \quad (9.51)$$

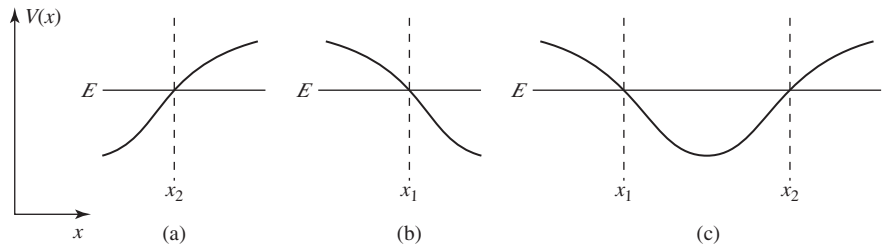


Figure 9.12: Upward-sloping and downward-sloping turning points.

In particular, if we’re talking about a potential *well* (Figure 9.12(c)), the wave function in the “interior” region ($x_1 < x < x_2$) can be written *either* as

$$\psi(x) \approx \frac{2D}{\sqrt{p(x)}} \sin \theta_2(x), \quad \text{where} \quad \theta_2(x) \equiv \frac{1}{\hbar} \int_x^{x_2} p(x') dx' + \frac{\pi}{4},$$

(Equation 9.47), or as

$$\psi(x) \approx \frac{-2D'}{\sqrt{p(x)}} \sin \theta_1(x), \quad \text{where} \quad \theta_1(x) \equiv -\frac{1}{\hbar} \int_{x_1}^x p(x') dx' - \frac{\pi}{4},$$

(Equation 9.51). Evidently the arguments of the sine functions must be equal, modulo π .¹³ $\theta_2 = \theta_1 + n\pi$, from which it follows that

$$\int_{x_1}^{x_2} p(x) dx = \left(n - \frac{1}{2}\right) \pi \hbar, \quad \text{with} \quad n = 1, 2, 3, \dots \quad (9.52)$$

This quantization condition determines the allowed energies for the “typical” case of a potential well with two sloping sides. Notice that it differs from the formulas for two vertical walls (Equation 9.17) or one vertical wall (Equation 9.48) only in the number that is subtracted from n (0, 1/4, or 1/2). Since the WKB approximation works best in the semi-classical (large n) regime, the distinction is more in appearance than in substance. In any event, the result is extraordinarily powerful, for it enables us to calculate (approximate) allowed energies *without ever solving the Schrödinger equation*, by simply evaluating one integral. The wave function itself has dropped out of sight.

**

Problem 9.6 The “bouncing ball” revisited. Consider the quantum mechanical analog to the classical problem of a ball (mass m) bouncing elastically on the floor.¹⁴

- What is the potential energy, as a function of height x above the floor? (For negative x , the potential is *infinite*—the ball can’t get there at all.)
- Solve the Schrödinger equation for this potential, expressing your answer in terms of the appropriate Airy function (note that $\text{Bi}(z)$ blows up for large z , and must therefore be rejected). Don’t bother to normalize $\psi(x)$.
- Using $g = 9.80 \text{ m/s}^2$ and $m = 0.100 \text{ kg}$, find the first four allowed energies, in joules, correct to three significant digits. *Hint:* See Milton Abramowitz and Irene A. Stegun, *Handbook of Mathematical Functions*, Dover, New York (1970), page 478; the notation is defined on page 450.
- What is the ground state energy, in eV, of an *electron* in this gravitational field? How high off the ground is this electron, on the average? *Hint:* Use the virial theorem to determine $\langle x \rangle$.

*

Problem 9.7 Analyze the bouncing ball (Problem 9.6) using the WKB approximation.

- Find the allowed energies, E_n , in terms of m , g , and $\hbar$.
- Now put in the particular values given in Problem 9.6(c), and compare the WKB approximation to the first four energies with the “exact” results.
- About how large would the quantum number n have to be to give the ball an average height of, say, 1 meter above the ground?

¹³ Not 2π —an overall minus sign can be absorbed into the normalization factors D and D' .

¹⁴ For more on the quantum bouncing ball see Problem 2.59, J. Gea-Banacloche, *Am. J. Phys.* **67**, 776 (1999), and N. Wheeler, “Classical/quantum dynamics in a uniform gravitational field”, unpublished Reed College report (2002). This may sound like an awfully artificial problem, but the experiment has actually been done, using neutrons (V. V. Nesvizhevsky *et al.*, *Nature* **415**, 297 (2002)).

- * **Problem 9.8** Use the WKB approximation to find the allowed energies of the harmonic oscillator.

Problem 9.9 Consider a particle of mass m in the n th stationary state of the harmonic oscillator (angular frequency ω).

- (a) Find the turning point, x_2 .
 (b) How far (d) could you go *above* the turning point before the error in the linearized potential (Equation 9.33, but with the turning point at x_2) reaches 1%? That is, if

$$\frac{V(x_2 + d) - V_{\text{lin}}(x_2 + d)}{V(x_2)} = 0.01,$$

what is d ?

- (c) The asymptotic form of $\text{Ai}(z)$ is accurate to 1% as long as $z \geq 5$. For the d in part (b), determine the smallest n such that $\alpha d \geq 5$. (For any n larger than this there exists an overlap region in which the linearized potential is good to 1% *and* the large- z form of the Airy function is good to 1%.)

- ** **Problem 9.10** Derive the connection formulas at a downward-sloping turning point, and confirm Equation 9.51.

- *** **Problem 9.11** Use appropriate connection formulas to analyze the problem of scattering from a barrier with sloping walls (Figure 9.13). *Hint:* Begin by writing the WKB wave function in the form

$$\psi(x) \approx \begin{cases} \frac{1}{\sqrt{p(x)}} \left[A e^{-\frac{i}{\hbar} \int_x^{x_1} p(x') dx'} + B e^{\frac{i}{\hbar} \int_x^{x_1} p(x') dx'} \right], & x < x_1; \\ \frac{1}{\sqrt{|p(x)|}} \left[C e^{\frac{1}{\hbar} \int_{x_1}^x |p(x')| dx'} + D e^{-\frac{1}{\hbar} \int_{x_1}^x |p(x')| dx'} \right], & x_1 < x < x_2; \\ \frac{1}{\sqrt{p(x)}} \left[F e^{\frac{i}{\hbar} \int_{x_2}^x p(x') dx'} \right], & x > x_2. \end{cases} \quad (9.53)$$

Do *not* assume $C = 0$. Calculate the tunneling probability, $T = |F|^2/|A|^2$, and show that your result reduces to Equation 9.23 in the case of a broad, high barrier.

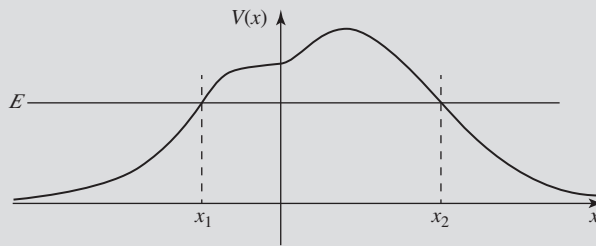


Figure 9.13: Barrier with sloping walls.



Problem 9.12 For the “half-harmonic oscillator” (Example 9.3), make a plot comparing the normalized WKB wave function for $n = 3$ to the exact solution. You’ll have to experiment to determine how wide to make the patching region. *Note:* You *can* do the integrals of $p(x)$ by hand, but feel free to do them numerically. You’ll need to do the integral of $|\psi_{\text{WKB}}|^2$ numerically to normalize the wave function.

FURTHER PROBLEMS ON CHAPTER 9

** **Problem 9.13** Use the WKB approximation to find the allowed energies of the general power-law potential:

$$V(x) = \alpha|x|^\nu,$$

where ν is a positive number. Check your result for the case $\nu = 2$. *Answer:*¹⁵

$$E_n = \alpha \left[(n - 1/2) \hbar \sqrt{\frac{\pi}{2m\alpha}} \frac{\Gamma\left(\frac{1}{\nu} + \frac{3}{2}\right)}{\Gamma\left(\frac{1}{\nu} + 1\right)} \right]^{\left(\frac{2\nu}{\nu+2}\right)}. \quad (9.54)$$

** **Problem 9.14** Use the WKB approximation to find the bound state energy for the potential in Problem 2.52. Compare the exact answer: $-\left[(9/8) - (1/\sqrt{2})\right] \hbar^2 a^2 / m$.

Problem 9.15 For spherically symmetrical potentials we can apply the WKB approximation to the radial part (Equation 4.37). In the case $l = 0$ it is reasonable¹⁶ to use Equation 9.48 in the form

$$\int_0^{r_0} p(r) dr = (n - 1/4) \pi \hbar, \quad (9.55)$$

where r_0 is the turning point (in effect, we treat $r = 0$ as an infinite wall). Exploit this formula to estimate the allowed energies of a particle in the logarithmic potential

$$V(r) = V_0 \ln(r/a)$$

(for constants V_0 and a). Treat only the case $l = 0$. Show that the spacing between the levels is independent of mass. *Partial answer:*

$$E_{n+1} - E_n = V_0 \ln\left(\frac{n + 3/4}{n - 1/4}\right).$$

¹⁵ As always, the WKB result is most accurate in the semi-classical (large n) regime. In particular, Equation 9.54 is not very good for the ground state ($n = 1$). See W. N. Mei, *Am. J. Phys.* **66**, 541 (1998).

¹⁶ Application of the WKB approximation to the radial equation raises some delicate and subtle problems, which I will not go into here. The classic paper on the subject is R. Langer, *Phys. Rev.* **51**, 669 (1937).

**

Problem 9.16 Use the WKB approximation in the form of Equation 9.52,

$$\int_{r_1}^{r_2} p(r) dr = (n' - 1/2) \pi \hbar \quad (9.56)$$

to estimate the bound state energies for hydrogen. Don't forget the centrifugal term in the effective potential (Equation 4.38). The following integral may help:

$$\int_a^b \frac{1}{x} \sqrt{(x-a)(b-x)} dx = \frac{\pi}{2} (\sqrt{b} - \sqrt{a})^2. \quad (9.57)$$

Answer:

$$E_{n'\ell} \approx \frac{-13.6 \text{ eV}}{[n' - (1/2) + \sqrt{\ell(\ell+1)}]^2}. \quad (9.58)$$

I put a prime on n' , because there is no reason to suppose it corresponds to the n in the Bohr formula. Rather, it orders the states *for a given ℓ* , counting the number of nodes in the radial wave function.¹⁷ In the notation of Chapter 4, $n' = N = n - \ell$ (Equation 4.67). Put this in, expand the square root ($\sqrt{1+\epsilon} = 1 + \frac{1}{2}\epsilon - \frac{1}{8}\epsilon^2 + \dots$), and compare your result to the Bohr formula.

Problem 9.17 Consider the case of a symmetrical double well, such as the one pictured in Figure 9.14. We are interested in bound states with $E < V(0)$.

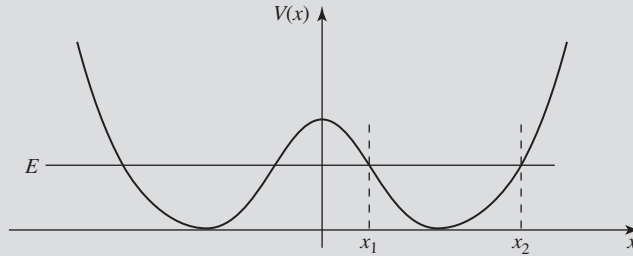


Figure 9.14: Symmetric double well; Problem 9.17.

- (a) Write down the WKB wave functions in regions (i) $x > x_2$, (ii) $x_1 < x < x_2$, and (iii) $0 < x < x_1$. Impose the appropriate connection formulas at x_1 and x_2 (this has already been done, in Equation 9.47, for x_2 ; you will have to work out x_1 for yourself), to show that

$$\psi(x) \approx \begin{cases} \frac{D}{\sqrt{|p(x)|}} \exp\left[-\frac{1}{\hbar} \int_{x_2}^x |p(x')| dx'\right], & \text{(i)} \\ \frac{2D}{\sqrt{p(x)}} \sin\left[\frac{1}{\hbar} \int_x^{x_2} p(x') dx' + \frac{\pi}{4}\right], & \text{(ii)} \\ \frac{D}{\sqrt{|p(x)|}} \left[2 \cos \theta e^{\frac{1}{\hbar} \int_x^{x_1} |p(x')| dx'} + \sin \theta e^{-\frac{1}{\hbar} \int_x^{x_1} |p(x')| dx'}\right], & \text{(iii)} \end{cases}$$

where

$$\theta \equiv \frac{1}{\hbar} \int_{x_1}^{x_2} p(x) dx. \quad (9.59)$$

¹⁷ I thank Ian Gatland and Owen Vajk for pointing this out.

- (b) Because $V(x)$ is symmetric, we need only consider even (+) and odd (−) wave functions. In the former case $\psi'(0) = 0$, and in the latter case $\psi(0) = 0$. Show that this leads to the following quantization condition:

$$\tan \theta = \pm 2e^\phi, \quad (9.60)$$

where

$$\phi \equiv \frac{1}{\hbar} \int_{-x_1}^{x_1} |p(x')| dx'. \quad (9.61)$$

Equation 9.60 determines the (approximate) allowed energies (note that E comes into x_1 and x_2 , so θ and ϕ are both functions of E).

- (c) We are particularly interested in a high and/or broad central barrier, in which case ϕ is large, and e^ϕ is *huge*. Equation 9.60 then tells us that θ must be very close to a half-integer multiple of π . With this in mind, write $\theta = (n + 1/2)\pi + \epsilon$, where $|\epsilon| \ll 1$, and show that the quantization condition becomes

$$\theta \approx \left(n + \frac{1}{2}\right)\pi \mp \frac{1}{2}e^{-\phi}. \quad (9.62)$$

- (d) Suppose each well is a parabola:¹⁸

$$V(x) = \begin{cases} \frac{1}{2}m\omega^2(x+a)^2, & x < 0, \\ \frac{1}{2}m\omega^2(x-a)^2, & x > 0. \end{cases} \quad (9.63)$$

Sketch this potential, find θ (Equation 9.59), and show that

$$E_n^\pm \approx \left(n + \frac{1}{2}\right)\hbar\omega \mp \frac{\hbar\omega}{2\pi}e^{-\phi}. \quad (9.64)$$

Comment: If the central barrier were *impenetrable* ($\phi \rightarrow \infty$), we would simply have two detached harmonic oscillators, and the energies, $E_n = (n + 1/2)\hbar\omega$, would be doubly degenerate, since the particle could be in the left well or in the right one. When the barrier becomes *finite* (putting the two wells into “communication”), the degeneracy is lifted. The even states (ψ_n^+) have slightly *lower* energy, and the odd ones (ψ_n^-) have slightly higher energy.

- (e) Suppose the particle starts out in the *right* well—or, more precisely, in a state of the form

$$\Psi(x, 0) = \frac{1}{\sqrt{2}}(\psi_n^+ + \psi_n^-),$$

which, assuming the phases are picked in the “natural” way, will be concentrated in the right well. Show that it oscillates back and forth between the wells, with a period

$$\tau = \frac{2\pi^2}{\omega}e^\phi. \quad (9.65)$$

¹⁸ Even if $V(x)$ is not strictly parabolic in each well, this calculation of θ , and hence the result (Equation 9.64) will be *approximately* correct, in the sense discussed in Section 2.3, with $\omega \equiv \sqrt{V''(x_0)}/m$, where x_0 is the position of the minimum.

- (f) Calculate ϕ , for the specific potential in part (d), and show that for $V(0) \gg E$, $\phi \sim m\omega a^2/\hbar$.

Problem 9.18 Tunneling in the Stark Effect. When you turn on an external electric field, the electron in an atom can, in principle, tunnel out, ionizing the atom. *Question:* Is this likely to happen in a typical Stark effect experiment? We can estimate the probability using a crude one-dimensional model, as follows. Imagine a particle in a very deep finite square well (Section 2.6).

- (a) What is the energy of the ground state, measured up from the bottom of the well? Assume $V_0 \gg \hbar^2/ma^2$. *Hint:* This is just the ground state energy of the *infinite* square well (of width $2a$).
- (b) Now introduce a perturbation $H' = -\alpha x$ (for an electron in an electric field $\mathbf{E} = -E_{\text{ext}}\hat{i}$ we would have $\alpha = eE_{\text{ext}}$). Assume it is relatively weak ($\alpha a \ll \hbar^2/ma^2$). Sketch the total potential, and note that the particle can now tunnel out, in the direction of positive x .
- (c) Calculate the tunneling factor γ (Equation 9.23), and estimate the time it would take for the particle to escape (Equation 9.29). *Answer:* $\gamma = \sqrt{8mV_0^3/3\alpha\hbar}$, $\tau = (8ma^2/\pi\hbar)e^{2\gamma}$.
- (d) Put in some reasonable numbers: $V_0 = 20$ eV (typical binding energy for an outer electron), $a = 10^{-10}$ m (typical atomic radius), $E_{\text{ext}} = 7 \times 10^6$ V/m (strong laboratory field), e and m the charge and mass of the electron. Calculate τ , and compare it to the age of the universe.

Problem 9.19 About how long would it take for a (full) can of beer at room temperature to topple over spontaneously, as a result of quantum tunneling? *Hint:* Treat it as a uniform cylinder of mass m , radius R , and height h . As the can tips, let x be the height of the center above its equilibrium position ($h/2$). The potential energy is mgx , and it topples when x reaches the critical value $x_0 = \sqrt{R^2 + (h/2)^2} - h/2$. Calculate the tunneling probability (Equation 9.23), for $E = 0$. Use Equation 9.29, with the thermal energy $((1/2)mv^2 = (1/2)k_B T)$ to estimate the velocity. Put in reasonable numbers, and give your final answer in years.¹⁹

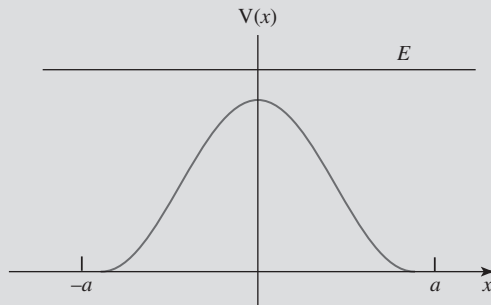


Figure 9.15: Reflection from a barrier (Problem 9.20).

¹⁹ R. E. Crandall, *Scientific American*, February, 1997, p. 74.

Problem 9.20 Equation 9.23 tells us the (approximate) transmission probability for tunneling through a barrier, when $E < V_{\max}$ —a classically forbidden process. In this problem we explore the complementary phenomenon: reflection from a barrier when $E > V_{\max}$ (again, a classically forbidden process). We'll assume that $V(x)$ is an even analytic function, that goes to zero as $x \rightarrow \pm\infty$ (Figure 9.15). *Question:* What is the analog to Equation 9.23?

- (a) Try the obvious approach: assume the potential vanishes for $|x| \geq a$, and use the WKB approximation (Equation 9.13) in the scattering region:

$$\psi(x) \begin{cases} = Ae^{ikx} + Be^{-ikx}, & x < a, \\ \approx \frac{1}{\sqrt{p(x)}} [C_+ e^{i\phi(x)} + C_- e^{-i\phi(x)}], & -a < x < a, \\ = Ce^{ikx}, & x > a. \end{cases} \quad (9.66)$$

Impose the usual boundary conditions at $\pm a$, and solve for the reflection probability, $R = |B|^2/|A|^2$.

Unfortunately, the result ($R = 0$) is uninformative. It's true that the R is exponentially small (just as the transmission coefficient is, for $E < V_{\max}$), but we've thrown the baby out with the bath water—this approximation is simply too drastic. The correct formula is

$$R = e^{-2\lambda}, \quad \text{where} \quad \lambda \equiv \frac{2}{\hbar} \int_0^{y_0} p(iy) dy \quad (9.67)$$

and y_0 is defined by $p(iy_0) = 0$. Notice that λ (like γ in Equation 9.23) goes like $1/\hbar$; it is in fact the leading term in an expansion in powers of $\hbar$: $\lambda = c_1/\hbar + c_2 + c_3\hbar + c_4\hbar^2 + \dots$. In the classical limit ($\hbar \rightarrow 0$), λ and γ go to infinity, so R and T go to zero, as expected. It is not easy to derive Equation 9.67,²⁰ but let's look at some examples.

- (b) Suppose $V(x) = V_0 \operatorname{sech}^2(x/a)$, for some positive constants V_0 and a . Plot $V(x)$, plot $p(iy)$ for $0 \leq y \leq y_0$, and show that $\lambda = (\pi a/\hbar) (\sqrt{2mE} - \sqrt{2mV_0})$. Plot R as a function of E , for fixed V_0 .
- (c) Suppose $V(x) = V_0/[1 + (x/a)^2]$. Plot $V(x)$, and express λ in terms of an elliptic integral. Plot R as a function of E .

²⁰ L. D. Landau and E. M. Lifshitz, *Quantum Mechanics: Non-Relativistic Theory*, Pergamon Press, Oxford (1958), pages 190–191. R. L. Jaffe, *Am. J. Phys.* **78**, 620 (2010) shows that *reflection* (for $E > V_{\max}$) can be regarded as *tunneling* in momentum space, and obtains Equation 9.67 by a clever analog to the argument yielding Equation 9.23.